

The positive unbounded product conjecture of arXiv:1401.5917 is answered by arXiv:1509.01571

Literature-status note for `fa_banach_001`

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Status. The conjecture is answered affirmatively by Gustafson and Mortad, Theorem 2.5 of arXiv:1509.01571. Their conclusion is stronger than the requested self-adjointness: it also identifies the closure of the reversed product.

Source conjecture (Chellali–Mortad, arXiv:1401.5917, Section 3, printed p. 8). *Let A and B be self-adjoint operators on a complex Hilbert space. Assume that B is bounded, $A \geq 0$, and AB is normal. Must AB be self-adjoint?*

The source explicitly says that neither a proof nor a counterexample was then known. Notice the domain asymmetry: $D(AB) = \{x : Bx \in D(A)\}$ need not equal $D(A) = D(BA)$, and BA need not be closed even when AB is normal.

Answering theorem (Gustafson–Mortad, arXiv:1509.01571, Theorem 2.5, printed pp. 5–8). *Let A and B be self-adjoint, let B be bounded, and suppose that $A \geq 0$ and AB is normal. Then AB and $\overline{BA}$ are self-adjoint, and*

$$AB = \overline{BA}.$$

This theorem has exactly the source conjecture’s hypotheses, so it gives a direct affirmative answer without an additional invertibility, spectral-gap, or closedness assumption on BA .

Idea of the answering proof. Because B is bounded and self-adjoint,

$$(BA)^* = AB, \quad \overline{BA} = (AB)^*.$$

Normality of AB therefore makes both AB and $\overline{BA}$ normal. Fuglede–Putnam applied to a suitable intertwining identity first yields $B^2A \subset AB^2$, and a square-root commutativity theorem gives $|B|A \subset A|B|$. Writing the polar decomposition $B = U|B|$ with $U = U^* = U^{-1}$, the authors establish the remaining intertwining relations and deduce

$$BA \subset AB.$$

Taking adjoints gives $(AB)^* \subset AB$. Since a normal operator and its adjoint have the same domain, this forces $AB = (AB)^*$; the identity $AB = \overline{BA}$ then follows from the first display.

Identification check. The later paper’s introduction restates two questions with $A \geq 0$ and B bounded self-adjoint: the AB -normal and BA -normal cases. It cites Chellali–Mortad directly, labels these as its main questions, and proves the AB case in Theorem 2.5. Its Proposition 2.14 also uses that theorem to answer the source paper’s related λ -commutativity problem.

References

- [1] C. Chellali and M. H. Mortad, *Commutativity up to a factor for bounded and unbounded operators*, arXiv:1401.5917.
- [2] K. Gustafson and M. H. Mortad, *Conditions Implying Commutativity of Unbounded Self-adjoint Operators and Related Topics*, arXiv:1509.01571; J. Operator Theory 76 (2016), 159–169.